CHAPTER 4

Boolean Algebra

Variables & operations · all the laws · De Morgan · consensus · duality · simplifying step by step

QUICK TRICK
Green boxes = quick tricks. Boolean algebra looks scary, but there are only about 10 rules. Learn the tricks and simplification becomes a game.

4.1 What is Boolean algebra?

● **LEVEL 1 · BEGINNER**

What is it? Normal algebra works with many numbers. **Boolean algebra works with only two values: 0 and 1.** We give a value a name like A, B or C. These names are called **variables**. Numbers 0 and 1 are called **constants**.

Why do we need it? Every digital circuit is a Boolean expression in disguise. If we can make the expression shorter, the circuit needs fewer gates. Fewer gates = cheaper, faster, less power.

Meaning of 0 and 1: 0 = FALSE = OFF = LOW. 1 = TRUE = ON = HIGH.

An **expression** joins variables using three operations: AND, OR and NOT.

Operation	Symbol	Simple meaning	Real-life example
AND	A · B or AB	Answer is 1 only if **both** are 1	To enter the exam hall you need admit card **AND** ID card
OR	A + B	Answer is 1 if **at least one** is 1	You can pay by cash **OR** card
NOT	A' (say “A bar”)	Flips the value: 0→1, 1→0	The opposite of ON is OFF

A	B	A · B (AND)	A + B (OR)	A' (NOT A)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

EASY MEMORY
AND is like multiplication (anything times 0 is 0). **OR is like addition, but 1 + 1 = 1** (not 2!). Only 0 and 1 exist.

QUICK TRICK
Order of work: N-A-O = **N**OT first, then **A**ND, then **O**R. So A + B·C' means A + (B·(C')). Use brackets to change the order.

4.2 The laws of Boolean algebra

● LEVEL 2 · EXAM

A **law** is a rule that is always true. Use these rules to make expressions shorter. Below, every law is written in simple words first.

Law	OR form	AND form	In simple words
Identity	A + 0 = A	A · 1 = A	Adding a 0 (or multiplying by 1) changes nothing
Null (Dominance)	A + 1 = 1	A · 0 = 0	A 1 in an OR wins. A 0 in an AND wins
Idempotent	A + A = A	A · A = A	Repeating a variable does nothing
Complement	A + A' = 1	A · A' = 0	A value and its opposite: OR gives 1, AND gives 0
Involution	(A')' = A		Two NOTs cancel each other
Commutative	A + B = B + A	AB = BA	Order does not matter
Associative	(A+B)+C = A+(B+C)	(AB)C = A(BC)	Grouping does not matter
Distributive	A(B + C) = AB + AC	**A + BC = (A+B)(A+C)**	The second form is special: it does NOT work in normal algebra
Absorption	A + AB = A	A(A + B) = A	The small term swallows the big term
Redundancy (extra)	A + A'B = A + B	A(A' + B) = AB	A complemented variable can be removed
De Morgan	(A + B)' = A' B'	(AB)' = A' + B'	Break the bar and change the sign
Consensus	AB + A'C + BC = AB + A'C	(A+B)(A'+C)(B+C) = (A+B)(A'+C)	The “consensus” term BC is not needed

COMMON MISTAKE

Treating Boolean algebra like normal algebra. Remember: A + A = **A** (not 2A), A · A = **A** (not A^2), A + 1 = **1**.

Truth table proofs (so you trust the laws)

If two expressions have the **same output on every row**, they are equal. Compare the last two columns.

Absorption: A + AB = A

A	B	AB	A + AB
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Redundancy: A + A'B = A + B

A	B	A'B	A + A'B	A + B
0	0	0	0	0
0	1	1	1	1
1	0	0	1	1
1	1	0	1	1

Column “A + A'B” and column “A + B” are identical, so the law is true.

De Morgan: (AB)' = A' + B' and **(A + B)' = A'B'**

A	B	(AB)'	A'+B'	(A+B)'	A'B'
0	0	1	1	1	1
0	1	1	1	0	0
1	0	1	1	0	0
1	1	0	0	0	0

Second distributive law: A + BC = (A + B)(A + C)

A	B	C	BC	A + BC	(A+B)(A+C)
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	1	1	1

Consensus: AB + A'C + BC = AB + A'C

A	B	C	AB	A'C	BC	AB+A'C+ BC	AB+A'C
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

4.3 Tricks to remember the big laws

De Morgan's law

QUICK TRICK
“Break the bar, change the sign.”
(A · B)' → break the bar into A' and B', change AND to OR → **A' + B'**
(A + B)' → break the bar into A' and B', change OR to AND → **A' · B'**
For 3 or more variables it works the same: (A + B + C)' = A' B' C' and (ABC)' = A' + B' + C'.

QUICK TRICK
Complement of a big expression — fast method. Swap every + with · (and · with +), and put a bar on every variable (remove the bar if it already has one).
Example: (A + B'C)' → A' · (B + C') = **A'(B + C')**.
Example: (AB' + C'D)' → (A' + B)(C + D').

COMMON MISTAKE
(A + B)' is **NOT** A' + B'. It is **A'B'**. The sign must change when the bar is broken.

Absorption

QUICK TRICK
“The small term swallows the big term.” If a short term (like A) appears, any longer term that *contains* it (like AB, ABC, ABD…) is not needed. A + AB = A. A + ABC = A. For the AND form: A(A + B) = A.

Redundancy (the most useful trick)

QUICK TRICK
“A + A'B = A + B.” When you see a variable and, in another OR term, the **opposite** of that variable multiplied by something, the opposite variable disappears.
Examples: B + B'C = B + C | X + X'Y = X + Y | A + A'BC = A + BC.
AND form: A(A' + B) = AB.

Consensus

QUICK TRICK
Consensus trick. Look for two terms where one has a variable A and the other has A' (like AB and A'C). Join the **other** parts (B and C) into the term **BC**. If BC is present in the expression, **cross it out**. It is not needed.
Example: AB + A'C + BC = AB + A'C. Also BCD (bigger than BC) is not needed either.
AND form: (A + B)(A' + C)(B + C) → (A + B)(A' + C).

Second distributive law

QUICK TRICK
A + BC = (A + B)(A + C). Think: “A can go inside both brackets.”
Use it to change a sum of products into a product of sums. Example: A + B'C = (A + B')(A + C).

4.4 Duality principle

● LEVEL 2 · EXAM

Dual of an expression: swap **AND ↔ OR** and swap **0 ↔ 1**. Keep the variables as they are. **Rule:** if a law is true, its dual is also true. That is why laws come in pairs (OR form and AND form).

Expression	Dual (swap + and ·, 0 and 1)	Complement (different thing!)
A + B	A · B	A' B'
A(B + C')	A + B C'	A' + B' C
A B + 0	(A + B) · 1	(A' + B') · 1
A + 1 = 1	A · 0 = 0 (also true)	—

QUICK TRICK

Dual = swap the operators and the 0/1, but keep the variables. Complement = also change the variables. Do not mix them up.

4.5 Simplifying expressions step by step

● LEVEL 2 · EXAM

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HOW TO SIMPLIFY  (follow this order)
1) Remove brackets and NOT-bars over big parts using De Morgan.
2) Look for a COMMON FACTOR (like A) in two terms -> take it out.
3) Look for X + X' = 1 or X . X' = 0.
4) Look for ABSORPTION  (A + AB = A).
5) Look for A + A'B = A + B.
6) Look for CONSENSUS.
7) Stop when none of the rules can be used.
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QUICK TRICK

Lazy check trick: after simplifying, put A = 0 and A = 1 (and the same for B, C) into both the original and the final expression. If the answers match for every combination, you are correct. (Or compare truth tables.)

Example 1 (easy): AB + AB'

Take the common factor A: A(B + B')

B + B' = 1, so A · 1 = **A**

Example 2: A + A'B + AB

A + A'B = A + B (redundancy trick). Now we have A + B + AB.

AB is swallowed by A (absorption). Answer: **A + B**

Example 3: (A + B)(A + C)

Multiply: AA + AC + BA + BC = A + AC + AB + BC (since AA = A)

A swallows AC and AB. Answer: **A + BC** (the second distributive law)

Example 4: A'B'C + A'BC + AB'C + ABC

Every term has C. Take C out: C(A'B' + A'B + AB' + AB)

Inside the bracket all four combinations of A and B appear, so it equals 1. Answer: **C**

Trick: if all combinations of some variables appear together, those variables disappear.

Example 5 (De Morgan): ((A + B)' + C)'

Break the outside bar: ((A + B)')' · C' → (A + B) · C'

Answer: **(A + B) C'**

Example 6 (consensus): AB + A'C + BC

B and C are the "other parts" of AB and A'C, and BC is present → cross it out.

Answer: **AB + A'C**

Example 7 (consensus, AND form): (A + B)(A' + C)(B + C)

Same idea: B + C is the consensus of (A + B) and (A' + C). Answer: **(A + B)(A' + C)**

Example 8 (Level 3): A'B + AB' + AB

A'B + AB = B(A' + A) = B. Now we have B + AB'.

B + AB' = A + B (redundancy trick with the roles swapped: B + B'A = B + A).

Answer: **A + B**

Example 9 (Level 3): (A + B)(A' + B)(A + B')

(A + B)(A' + B) = B + AA' = B + 0 = B (second distributive law).

B · (A + B') = AB + BB' = AB + 0 = **AB**

4.6 Common mistakes and exam traps

COMMON MISTAKE

- (A + B)' written as A' + B' (wrong). Correct: A'B'.
- A + A = 2A (wrong). Correct: A.
- (AB)' written as A'B' (wrong). Correct: A' + B'.
- Thinking A + BC = (A + B) + C. It is (A + B)(A + C).
- Forgetting a bar when you apply De Morgan.

EXAM TRAP

GATE often asks: "which of these expressions is equal to F?". Do not simplify everything. **Test with numbers**: put A, B, C = 0 or 1 and check which option matches. It is faster.

GATE POINT

Consensus theorem and **De Morgan** are the two most tested laws. Learn both forms (OR form and AND form).

Quick-Solve Tricks — Summary

Situation	Trick
See a bar over a big part	Break the bar, change the sign (AND ↔ OR), bar every variable.
X + X·(something)	Absorption: the short term swallows the long term.
X + X'·Y	Becomes X + Y. The X' disappears.
X·(X' + Y)	Becomes X·Y.
AB + A'C (+ BC)	BC is the consensus. Cross it out.
All combinations of a variable together	That variable disappears (A + A' = 1).
Two terms differ in one variable	Factor and remove that variable: AB + AB' = A.
Need the dual	Swap + ↔ · and 0 ↔ 1. Keep variables.
Not sure if two expressions are equal	Try all 0/1 values (truth table).
Order of operations	N-A-O: NOT, then AND, then OR.

GATE QUICK CHECK — Chapter 4

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Identity  : A + 0 = A      A . 1 = A          Null      : A + 1 = 1      A . 0 = 0
Idempotent: A + A = A      A . A = A          Complement: A + A' = 1     A . A' = 0
Absorption: A + AB = A     A(A + B) = A       Redundancy: A + A'B = A + B
De Morgan : (A+B)' = A'B'  (AB)' = A' + B'    Distributive: A + BC = (A+B)(A+C)
Consensus : AB + A'C + BC = AB + A'C          (A+B)(A'+C)(B+C) = (A+B)(A'+C)
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Traps: De Morgan sign change · dual is not complement · 1 + 1 = 1 · second distributive law · consensus term may be hidden inside a bigger term.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Simplify (A + 1) · B.

Q2. What is (A')' + A · 1?

Q3. Write the dual of A + B · 1.

● **LEVEL 2 · EXAM**

Q4. Simplify AB + AB'C.

Q5. Find the complement of A + B'C.

Q6. Simplify A'B + A'B' + AB'.

● **LEVEL 3 · GATE**

Q7. Simplify AB + A'C + BCD.

Q8. Simplify (A + C)(A' + B)(B + C).

Q9. True or false: A + BC = (A + B)(A + C).

Q10. True or false: (A + B)' = A' + B'.

Q11. Simplify (A + B + C)(A + B + C')(A + B' + C).

Q12. How many literals are in the simplest form of A'B + AB' + AB ? (A literal is one variable, with or without a bar.)

Answers

Q	Solution
1	A + 1 = 1 (null law). 1 · B = **B**.
2	(A')' = A (involution), A · 1 = A. A + A = **A**.
3	Swap + and ·, and 1 becomes 0: **A · (B + 0)**.
4	Take A out: A(B + B'C). Redundancy trick: B + B'C = B + C. Answer **A(B + C) = AB + AC**.
5	Fast method: swap operators and bar the variables. A' · (B + C'). Answer **A'(B + C')**. (Check with De Morgan: (A + B'C)' = A' · (B'C)' = A'(B + C').)
6	A'B + A'B' = A'(B + B') = A'. Then A' + AB' = A' + B' (redundancy). Answer **A' + B'**.
7	A and A' are in AB and A'C. Consensus is BC. BCD contains BC, so it is not needed. Answer **AB + A'C**.
8	Rename: (A + C)(A' + B)(C + B) has the shape (A + X)(A' + Y)(X + Y) with X = C, Y = B. The last bracket is the consensus. Answer **(A + C)(A' + B)**.
9	**True** (second distributive law; proof by truth table above).
10	**False.** Correct: (A + B)' = A'B'.
11	(A + B + C)(A + B + C') = A + B (the C variable disappears: (A+B) + CC'). Then (A + B)(A + B' + C) = A + B(B' + C) = A + BC. Answer **A + BC**. (Check with A=0, B=1, C=0: the third bracket A + B' + C = 0 + 0 + 0 = 0, so the original is 0. And A + BC = 0 + 0 = 0. ✔)
12	A'B + AB = B. B + AB' = A + B. Answer A + B has **2 literals**.

Next: Chapter 5 — Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR, buffer, universal gates, and quick tricks for building circuits).